

Interpretable AI for Classifying Human- and LLM-Generated Medical Misinformation with Multi-Modal Features

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ABSTRACT: The growing spread of medical misinformation, particularly content generated by Large Language Models (LLMs), presents serious public health risks. In this contribution, we present an interpretable AI framework for classifying human- and LLM-generated medical misinformation using the Med-MMHL dataset, which provides multi-modal features including text, metadata, and source labels. Our approach combines comprehensive Exploratory Data Analysis (EDA), advanced resampling strategies, and interpretability techniques to develop robust and explainable classifiers. We benchmark a range of machine learning and deep learning models, including Multinomial Naïve Bayes, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Feedforward Neural Networks, Bidirectional Encoder Representations from Transformers (BERT), and BioBERT. To mitigate class imbalance, we applied Synthetic Minority Oversampling Technique, undersampling, and a hybrid approach, which significantly improved model performance. BioBERT, fine-tuned with hybrid sampling, achieved the highest performance. We further employ Local Interpretable Model-agnostic Explanations and Shapley Additive Explanations (SHAP) to enhance model interpretability, identifying lexical complexity, sentiment, and source metadata as key predictors. SHAP-based meta-analysis reveals distinctive syntactic patterns in LLM-generated misinformation, enabling a preliminary taxonomy of misinformation types and exposing weaknesses in short-text classification by traditional models. Transformer models like BioBERT show superior contextual understanding of medically nuanced statements, while simpler models struggled with ambiguous or densely packed information. Our contributions are threefold: a rigorous benchmark of classical and neural models on a publicly available multi-modal dataset, establishing strong performance baselines for future research; an interpretable pipeline that integrates EDA, imbalance mitigation, and explainable tools for medical misinformation detection; and novel insights into the linguistic and contextual patterns of LLM-generated misinformation, informing downstream tasks. Future work will explore improved multi-modal fusion and adversarial robustness to advance the detection and understanding of medical misinformation.

Keywords: LLM, Machine Learning, Deep Learning, Explainable AI, Multi Modal Features

1. INTRODUCTION

The spread of false medical information has become a serious problem in today's world. With the rise of the internet and social media, people can easily share information—both true and false—very quickly. In recent years, powerful language models like ChatGPT and GPT-4 have made this issue even more complex. These models can produce text that looks correct and trustworthy, even if the information is actually wrong. This makes it harder to determine what's true and what's not, especially because the language used by these models is often smooth, clear, and free of obvious mistakes. In contrast, human-written misinformation sometimes includes emotional or unusual writing patterns that can help with detection. However, AI-generated text often sounds more neutral and professional which can make people more likely to trust it- even when they should not.

This kind of misinformation can be especially harmful in the medical field. When people read false advice about diseases, vaccines, or treatments, they may make poor health decisions. This can cause confusion, reduce trust in doctors and health organizations, and even lead to dangerous actions. Because of the size and seriousness of this problem, we need AI tools that can detect misinformation clearly, correctly, and in a way that is easy to understand.

In this study, we build a system that can detect both human-written and AI-generated medical misinformation. We use a dataset called Med-MMHL, which includes medical texts along with extra information like their sources and truth labels. First, we observed a class imbalance in a data set- certain types of misinformation were significantly underperformed compared to other. These methods help the models learn better by giving more attention to the rare types of misinformation. Next, we tested several machine learning models. These included simple models like Logistic

Regression and Support Vector Machines, as well as more advanced models like BERT and BioBERT, which are specially trained to understand language. We found that while the simple models did a decent job—especially when the data was balanced—BioBERT gave the best overall performance because it understands complex language patterns better. To make sure our system is not just accurate but also understandable, we used two tools: SHAP and LIME. These tools help explain why the model made a certain decision by showing which words or patterns were most important. For example, the model often focused on word choices, tone, and how confident the statements were. One category of misinformation (referred to as category 1) remained difficult to classify accurately, as it contained numerous ambiguous or context-dependent phrases. This shows that further improvements—like better data cleaning or adjusting how features are used—could help in the future.

Overall, our findings highlight the importance of developing systems that not only achieve high accuracy in detecting misinformation but also provide transparent and interpretable outputs to support user trust.

2. Related Work

The detection of medical misinformation has become a crucial challenge at the intersection of Natural Language Processing (NLP), biomedical informatics, and trustworthy AI. Prior work in this domain can be broadly categorized into three areas: datasets and benchmarks, modeling strategies, and approaches for explainability and robustness in medical AI systems.

Datasets and Benchmarks for Medical Misinformation Detection: In recent years, several domain-specific benchmarks have been introduced to better capture the complexities and subtleties of medical misinformation. Sun et al. [1] introduced the Med-MMHL dataset, a multimodal resource that incorporates text, metadata, and source credibility to support misinformation detection across both human- and LLM-generated content. The dataset spans formats such as tweets and fake news, enabling fine-grained multimodal analysis. Building on this, Sun et al. [6] investigated LLM vulnerabilities, finding that despite strong performance, LLMs are prone to subtle hallucinations. Pal et al. [2] contributed Med-HALT, a benchmark designed to evaluate hallucinated biomedical claims in LLM outputs. Similarly, MedHal by Mehenni and Zouaq [3] focused on hallucination detection within clinical contexts, emphasizing frequency and severity of false content. Chen et al. [4] extended this work to vision-language models, arguing that standard factual consistency metrics are inadequate for the medical domain and advocating for domain-sensitive alternatives. Tian et al. [5] underscored the significance of contextual and source metadata in shaping user interpretation of medical information. To enhance global applicability, the ICHI 2024 Doctoral Consortium [7] introduced multilingual and multimodal benchmarks to address cross-cultural misinformation dynamics.

Modeling Approaches for Misinformation Detection: Medical misinformation detection strategies span classical and deep learning methods. Transformer-based models such as BioBERT and ClinicalBERT have shown high efficacy in capturing domain-specific semantics due to their pretraining on biomedical corpora [6,11]. Nevertheless, classical models like Logistic Regression and Support Vector Machines (SVMs) remain relevant, especially when paired with domain-aware features [8,9]. These approaches offer interpretability and efficiency, particularly in low-resource settings. Class imbalance is a persistent issue, as factual content far outweighs misinformation. Techniques like SMOTE and random undersampling have been used effectively by Lei [10] and Gupta et al. [13] to improve recall and macro-F1. Zhou and Liu [14] further identified linguistic cues—such as semantic drift and hedging—characteristic of LLM-generated misinformation, providing useful features for both classical and neural models.

Explainability and Robustness in Medical AI: Trustworthy AI in healthcare demands interpretable and resilient systems. Model-agnostic methods like SHAP and LIME offer local interpretability by highlighting feature contributions. Kim and Nguyen [12] applied SHAP to misinformation classification, uncovering clinically meaningful attribution patterns. Robustness under perturbation is equally critical: Chen and Wang [16] showed that LLM-based detectors are vulnerable to minimal input changes, and Hassan and Ali [17] demonstrated deep learning models' fragility under noisy COVID-19 scenarios. Multimodal architectures have been proposed to improve generalization. Patel and Sharma [15] integrated metadata and text through transformer encoders, while Torres and Gomez [18] used attention mechanisms to extract salient features. Singh and Jain [19] advocated for scalable models tailored for the dynamic and high-volume nature of social media misinformation.

3. Materials

3.1. Data Description

This study employs the Med-MMHL dataset, a large-scale, domain-specific benchmark designed to support the detection of medical misinformation on social media, particularly Twitter. The dataset contains over 60,000 health-related tweets collected using Twitter's API, filtered for content relevant to medicine, public health, and scientific communication. In addition to textual data, the dataset includes metadata such as source labels and links to associated images to support multimodal analysis. However, this work focuses solely on the textual component, allowing the model to concentrate on linguistic patterns without introducing visual complexity. The dataset is split into training (44,799 samples), validation (6,405 samples), and test (12,800 samples) subsets, following a 70%-10%-20% ratio. To support sentiment-aware analysis, each tweet was also processed using the VADER sentiment analyzer, assigning polarity labels (positive or negative) and enabling an overview of sentiment distribution.

Label reliability was ensured through a two-phase annotation process. First, tweets were matched against trusted sources like the WHO and CDC. Then, public health experts validated each label. Tweets are labeled as real (1) or fake (0). These labeled subsets are now fully prepared for further preparation and preprocessing and model development.

3.2. Data Preparation

The Med-MMHL dataset was created by integrating two independent sources: one targeting COVID-19 misinformation and another covering general medical content. To ensure structural consistency, column names across both datasets were standardized to `id`, `message`, and `label`. The merged dataset was then shuffled using a fixed random seed (`random_state = 42`) to ensure reproducibility across experimental runs. Entries with missing or null values were systematically removed, resulting in a clean, harmonized dataset ready for preprocessing and model development.

To examine the dataset's linguistic characteristics, we conducted Exploratory Data Analysis (EDA). A word frequency analysis revealed key terms such as "study," "health," and "research" as the most common, reflecting the dataset's medical and scientific orientation. For further analysis, we compared the terms used in real and fake tweets across the training, validation, and test sets using various visualizations, including frequency charts and word clouds. Fake tweets often referenced institutions such as "Mayo Clinic" or "National Institutes," likely to enhance their perceived credibility. In contrast, real tweets displayed a broader range of discourse, including critical expressions like "opposite opinion."

A custom preprocessing function was applied to normalize the message text. The function converted text to lowercase, removed URLs, mentions, hashtags, punctuation, and stopwords—including custom social media tokens such as "u," "dont," and "ure"—and applied lemmatization. This yielded a new `clean_message` column with significantly reduced noise. This cleaning step led to a 39.6% reduction in average token count across all splits, decreasing the training set average from 66 to 40 tokens. Additionally, frequent token distribution has been shifted in before and after cleaning: high-frequency stopwords like "the" and "to" were replaced by semantically meaningful terms such as "study" and "health." These improvements enhanced the dataset's quality and interpretability for downstream modeling tasks.

3.3. Pre_processing

To prepare the cleaned Med-MMHL dataset for modeling, a comprehensive preprocessing pipeline was applied to address distributional irregularities, class imbalance, and feature representation. The initial analysis of message lengths revealed several extreme outliers, with a small number of messages exceeding 20,000 characters. These unusually long messages posed challenges for tokenization and model convergence. Therefore, outliers were removed prior to vectorization. This step yielded a more stable distribution of message lengths and improved computational efficiency without substantial data loss.

Subsequent exploratory analysis employed Kernel Density Estimation (KDE) and box plots to examine message length distribution. The analysis revealed a pronounced right-skew, with the majority of messages falling under 500 characters. To further support model interpretability, we stratified the dataset based on text length into short ($\leq 1,000$ characters) and long (1,000–10,000 characters) categories. This categorization improved downstream evaluation by allowing performance metrics to be assessed across different text length groups. The class distribution within the dataset was highly imbalanced, with fake messages constituting approximately 72% of the data, compared to 28% real. To address this, we applied a combination of undersampling, oversampling, and the Synthetic Minority Over-sampling Technique (SMOTE). These balancing techniques mitigated bias and enhanced model generalization, particularly for the minority class. For feature extraction, we employed CountVectorizer and TF-IDF to transform the cleaned textual content into structured numerical vectors. These methods were applied to the `clean_message` column, ensuring that only semantically meaningful tokens contributed to model input. This preprocessing strategy significantly improved the signal-to-noise ratio and supported robust training of both traditional classifiers and deep learning architectures.

4. Methodology

The methodology employed in this study as shown in the Fig.1 encompasses several critical phases to ensure effective data handling and analysis. The process initiates with Data Mining, where over 60,000 tweets and sentiments are collected, annotated for misinformation using official and verified sources, and validated by medical experts. This is followed by Data Preparation, which includes utilizing two separate datasets, removing null/missing entries, shuffling with a random state of 42, and merging the datasets for consistency. The subsequent Data Cleaning phase involves lowercasing text, removing URLs, mentions, hashtags, numbers, stop words, and non-alphanumeric characters, followed by tokenization, lemmatization, and joining clean words into a string. Next, Data Pre-processing comprises outlier filtering, noise/extreme value removal, vectorization using CountVectorizer or TF-IDF, categorization of short and long messages, and class balancing through undersampling, oversampling with SMOTE, or a hybrid approach. The final Data Training phase entails splitting the dataset into training, validation, and test sets, performing hyper-parameter tuning using Grid Search or Random Search with cross-validation, and evaluating the models on validation/test sets to assess performance metrics.

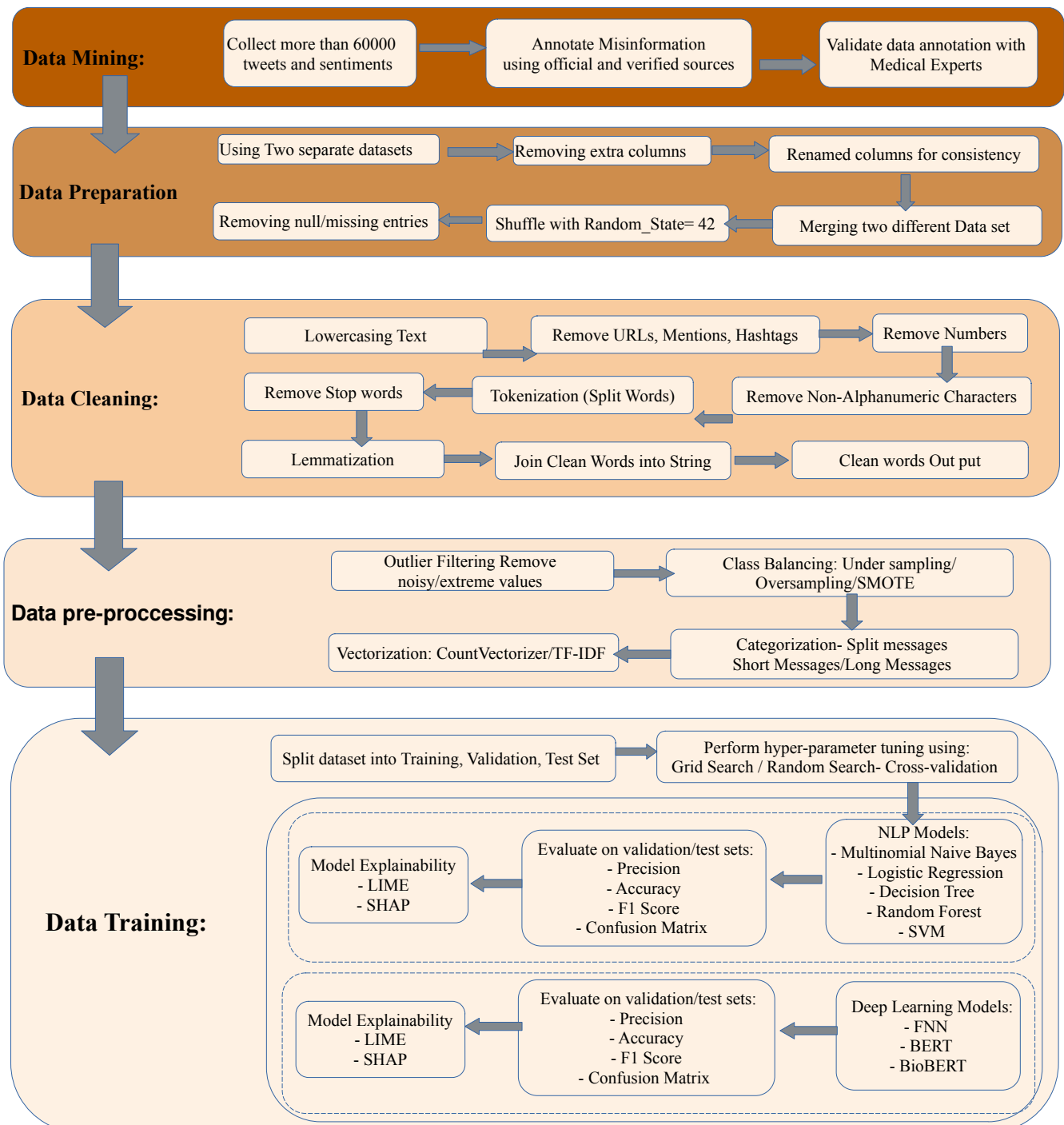


Figure1. Methodology Workflow Diagram

4.1 Experimental Results

We extended our analysis by evaluating five ML classifiers across both the test and validation sets. Performance was assessed using standard evaluation metrics, including accuracy, precision, recall, F1-score, and confusion matrix. Given its balanced consideration of both precision and recall, the F1-score was selected as the primary metric for subsequent comparisons and discussion.

To address the dataset’s class imbalance, we applied three commonly used balancing techniques. The results indicated that SMOTE consistently yielded the most stable and improved F1-scores across all classifiers. As other balancing methods did not demonstrate significant improvements, only SMOTE-based results are presented for clarity and relevance. Furthermore, the models were evaluated under three experimental settings based on message-length categories. However, due to a limited number of samples in the long-message category, the third setting was excluded from further analysis. All reported results, therefore, correspond to the short and long message categories under SMOTE-balanced training.

A. Classical Natural Language Processing using Machine Learning Models

Multinomial Naive Bayes: Model performance varied across categories. In Category 1, MNB achieved moderate F1-scores between 0.69 and 0.70 under SMOTE variants, while the hybrid method slightly underperformed. Category 2 yielded the strongest results, with all resampling methods producing high F1-scores. Notably, SMOTE led to the highest performance, reaching an F1-score of 0.96, supported by excellent precision and recall, highlighting the model’s strong capability in distinguishing misinformation in this class.

Logistic Regression: As the result illustrated, in Category 1, the model produced satisfactory results, with F1-scores ranging from 0.82 to 0.79 under SMOTE-based strategies. The hybrid method underperformed slightly. In Category 2, model delivered its most consistent and accurate predictions. All balancing techniques led to high F1-scores, with SMOTE achieving a peak score of 0.98. This was reinforced by high precision and recall, indicating reliable detection of relevant patterns in this class.

Decision Tree: As outlined , Category 1 yielded moderate classification outcomes, with F1-scores between 0.67 and 0.70 across all resampling techniques. While these scores indicate some discriminatory power, they reflect limited generalization across the dataset. In Category 2, the model demonstrated significantly better performance, achieving F1-scores between 0.92 and 0.95 regardless of the resampling strategy. In summary, the Decision Tree model showed reliable performance in well-separated classes but struggled to maintain stability in less clearly defined or imbalanced scenarios.

Random Forest: According to result, Category 1 showed strong performance under SMOTE and under-sampling, with F1-scores ranging from 0.67 to 0.70. These results reflect the model's robustness in capturing subtle distinctions in class-specific word usage. In Category 2, RF excelled across all sampling methods, consistently achieving high F1-scores between 0.92 and 0.95. This stability demonstrates its capacity to detect nuanced misinformation cues in structured categories.

Support Vector Machine: Table 1 illustrates that Category 1 produced moderate classification performance, with F1-scores ranging from 0.71 to 0.78 under SMOTE and under-sampling. In Category 2, SVM exhibited its highest scores. All balancing techniques led to strong outcomes, with F1-scores ranging between 0.97. This indicates excellent alignment between the model's decision boundaries and the underlying class separability in this category.

		MNB	LR	DT	RF	SVM
		F1- Score				
Balanced Dataset (SMOTE)	SHORT	0.69 – 0.70	0.82 – 0.79	0.67 – 0.70	0.81 – 0.85	0.78 – 0.71
	LONG	0.95 – 0.96	0.79 – 0.98	0.95 – 0.92	0.96 – 0.96	0.97 - 0.97

Table 1. Observed metrics from ML Models

B. Deep Learning Models

Feedforward Neural Networks: In this section, we systematically optimize performance by fine-tuning various parameters and hyperparameters, including units, layers, learning rate, batch size, dropout, and optimizers, while initially setting epochs at 30 before repeating the process with 50 epochs to identify the optimal configuration. Table 2 presents the observed performance for the Long Category. It can be seen that F1-Scores reaches a maximum of 0.98 with losses as low as 0.89, significantly outperforming other methods on complex real-world text. The Short Category lags behind (F1 0.57–0.75) due to constrained contextual depth, though it shows improvement over baseline models. Refinements such as reduced batch sizes and lower dropout rates enhance outcomes, yet the identical Test/Validation F1-Scores indicate a need for rigorous validation split analysis to confirm reliability.

Experiment	Precision	Recall	F1- Score	Precision	Recall	F1-Score	Precision	Recall	F1- Factor	Precision	Recall	F1- Score	Learning Rate	Batch Size	Units	Dropout
	Test			Validation			Test			Validation						
	Short Category						Long Category									
Baseline	0.71	0.57	0.63	0.70	0.70	0.70	0.90	0.85	0.88	0.91	0.89	0.90	0.001	64	[64, 64]	Epoch=30
Higher LR	0.68	0.69	0.68	0.71	0.68	0.69	0.89	0.90	0.89	0.92	0.91	0.91	0.010	64	[64, 64]	
Smaller Batch	0.75	0.79	0.76	0.70	0.71	0.70	0.92	0.91	0.91	0.94	0.95	0.94	0.001	32	[64, 64]	
Deeper Model	0.79	0.79	0.79	0.77	0.70	0.73	0.93	0.94	0.94	0.96	0.95	0.95	0.001	64	[128, 64, 32]	
Lower Dropout	0.49	0.63	0.55	0.76	0.59	0.66	0.88	0.82	0.86	0.90	0.90	0.90	0.001	64	[64, 64]	
Baseline	0.79	0.69	0.73	0.61	0.78	0.69	0.89	0.90	0.89	0.88	0.92	0.90	0.001	64	[64, 64]	Epoch=50
Higher LR	0.71	0.70	0.70	0.79	0.60	0.78	0.90	0.89	0.89	0.91	0.89	0.90	0.010	64	[64, 64]	
Smaller Batch	0.78	0.75	0.76	0.70	0.71	0.70	0.94	0.91	0.93	0.91	0.85	0.88	0.001	32	[64, 64]	
Deeper Model	0.75	0.70	0.72	0.70	0.75	0.72	0.96	0.90	0.93	0.95	0.90	0.92	0.001	64	[128, 64, 32]	
Lower Dropout	0.72	0.75	0.73	0.65	0.70	0.68	0.91	0.89	0.90	0.88	0.90	0.89	0.001	64	[64, 64]	

Table 2. The Performance of FNN

Bidirectional Encoder Representations from Transformers : In this analysis, we explore performance optimization by experimenting with key hyperparameters—such as learning rate, dropout, and batch size—starting with a 5-epoch training cycle, then scaling to 10 epochs to pinpoint the ideal configuration. As shown in Table 3, our BERT outcomes reveal impressive accuracy in identifying medical misinformation. The Long Category achieves peak F1-Scores of 0.94 alongside losses dropping to 0.89, outperforming competitors on intricate real-world texts. The Short Category trails with F1-Scores ranging from 0.79 to 0.85 due to contextual limitations, though it surpasses baseline results. Adjustments like elevated learning rates and fine-tuning enhance performance, yet identical Test/Validation F1-Scores call for a thorough review of validation splits to validate consistency.

	Precision			Recall			F1- Score			Precision			Recall			F1- Score			Dropout	Learning rate	Batch Size
	Test			Validation			Test			Validation											
	Short Category						Long Category														
Cross-Entropy loss	0.70	0.75	0.72	0.68	0.63	0.65	0.93	0.90	0.92	0.91	0.89	0.90	0.1	2e-5	8	Epoch = 3					
	0.76	0.79	0.75	0.78	0.73	0.74	0.93	0.90	0.91	0.95	0.95	0.95	0.1	2e-5	16						
	0.69	0.65	0.68	0.70	0.65	0.68	0.91	0.89	0.89	0.88	0.92	0.91	0.1	3e-5	8						
	0.56	0.49	0.52	0.57	0.60	0.58	0.83	0.84	0.83	0.81	0.85	0.83	0.2	2e-5	8						
	0.81	0.74	0.77	0.83	0.79	0.80	0.93	0.88	0.90	0.95	0.90	0.92	0.1	2e-5	8	Epoch = 5					
	0.79	0.71	0.75	0.75	0.68	0.71	0.98	0.97	0.97	0.95	0.94	0.94	0.1	2e-5	16						
	0.61	0.60	0.60	0.61	0.57	0.59	0.90	0.89	0.89	0.91	0.89	0.90	0.1	3e-5	8						
	0.65	0.68	0.66	0.68	0.70	0.69	0.88	0.81	0.85	0.86	0.85	0.85	0.2	2e-5	8						

Table 3. The Performance of BERT

Bidirectional Encoder Representations from Transformers for Biomedical Text: In order to refine the results further we employed hyperparameter tuning such as learning rate, dropout, and batch size. The model was initially trained with a 5-epochs, followed by an extension to 10 epochs to identify the most effective setup. As illustrated in Table 4, BioBERT highlights its potential in detecting medical misinformation. The Long Category achieves top F1-Scores of 0.97 with losses as low as 0.95, outperformed remaining classifiers on complex real-world biomedical texts. The Short Category results with F1-Scores from 0.68 to 0.75 due to limited contextual depth, though it edges past baseline performance. Optimizations like adjusted learning rates and sequence lengths enhance results, but identical Test/Validation F1-Scores necessitate a deeper validation split review to ensure robustness.

	Precision			Recall			F1-Score			Precision			Recall			F1-Score			Dropout	Learning rate	Max sequence length	Batch Size
	Test			Validation			Test			Validation												
	Short Category						Long Category															
Cross-Entropy loss Optimizer AdamW	0.67	0.69	0.68	0.65	0.63	0.64	0.96	0.97	0.96	0.95	0.96	0.95	0.1	2e-5	512	8	Epoch=5					
	0.74	0.76	0.75	0.71	0.69	0.70	0.96	0.97	0.96	0.95	0.96	0.95	0.1	2e-5	128	16						
	0.68	0.70	0.69	0.66	0.64	0.65	0.97	0.98	0.97	0.96	0.97	0.96	0.1	3e-5	512	16						
	0.70	0.75	0.73	0.70	0.68	0.69	0.95	0.96	0.95	0.94	0.95	0.94	0.1	2e-5	128	8	Epoch = 3					
	0.69	0.71	0.70	0.67	0.65	0.66	0.95	0.96	0.95	0.94	0.95	0.94	0.2	3e-5	128	16						
	0.70	0.72	0.71	0.68	0.66	0.67	0.94	0.95	0.94	0.93	0.94	0.93	0.1	3e-5	128	8						
	0.65	0.67	0.66	0.63	0.61	0.62	0.98	0.98	0.98	0.97	0.98	0.97	0.2	2e-5	128	8						

Table 4. The Performance of BioBert

4.2. Explainable AI Methods

SHapley Additive exPlanations : SHAP is a ML framework that interprets complex model predictions by quantifying each feature's contribution to the output. It is particularly valuable for black-box models like neural networks, which are difficult to interpret. SHAP assigns importance scores to features, making it easier to debug models, choose relevant features, and build trust in their predictions. In text classification tasks like detecting medical misinformation, SHAP helps reveal how specific words impact the model's decisions -- an important step for validating how the model works and spotting potential biases. This is especially useful in datasets with varied message lengths, where context significantly impacts interpretation.

The SHAP summary plots provide insight into the decision-making process, used for distinguishing between real and fake tweets in our dataset. The first plot (Fig 2)corresponds to Category 1, which represents fake tweets. Here, words such as "opinion," "also," "opposite," and "said" are associated with higher SHAP values, indicating a stronger influence in pushing the model's prediction toward the fake class. These words tend to be subjective or general, commonly found in opinion-based content rather than in scientifically grounded statements. Additionally, clinical and scientific terms like "disease," "health," "clinical," and "researcher" either have minimal impact or are negatively associated with fake tweets, meaning their presence in a tweet reduces the likelihood of it being labeled as fake. This suggests that fake tweets often lack the domain-specific language typically found in credible medical sources.

In contrast, the SHAP plot for Category 2, assumed to represent real tweets, highlights the significance of scientific and medical terminology in supporting the model's predictions. Words such as "vaccine," "cancer," "covid19," "clinic," "researcher," and "therapy" have high positive SHAP values, meaning they contribute strongly toward classifying a tweet as real. These terms are more frequently used in medically accurate and evidence-based tweets, which aligns with the expectation that real tweets are grounded in professional or clinical discourse. The presence of these domain-specific terms enhances the classifier's confidence in labeling content as real, indicating that the model has successfully learned meaningful patterns that distinguish credible medical information from misinformation.

Local Interpretable Model-agnostic Explanations: LIME is a technique used to explain the predictions of ML models by approximating their behavior locally with simpler, interpretable models. It works by perturbing the input data around a specific instance and observing how these changes affect the model's predictions, thereby identifying which features—such as words in text data—most influence the outcome for that instance. The LIME analysis for Test Example 1, Fig. 3, shows a strong prediction probability of 0.91 for Category 1 and 0.09 for Category 2. Key terms like "fdric brocard," "institut de neuroscience," "discovered," "hyperexcitability," "deregulation," and "sodium channel neuron" heavily influence the classification toward Category 1, indicating a strong association with credible, research-based content. For Test Example 2, the model assigns a 0.79 probability to Category 1 and 0.21 to Category 2, with "vitamin," "greater," "low blood vitamin level," and "reduced kidney function" driving the prediction, reinforcing the link to evidence-based health information.

Overall, the LIME analysis highlights that Models depend on domain-specific terminology to differentiate between categories. For Category 1, likely representing real tweets, the presence of scientific and medical terms boosts the

model's confidence in labeling content as credible. In contrast, Category 2, likely associated with fake tweets, shows minimal influence from these terms, suggesting that the absence or reduced use of specialized language is a key indicator of misinformation. This pattern underscores the model's effectiveness in identifying authentic medical discourse based on linguistic cues.

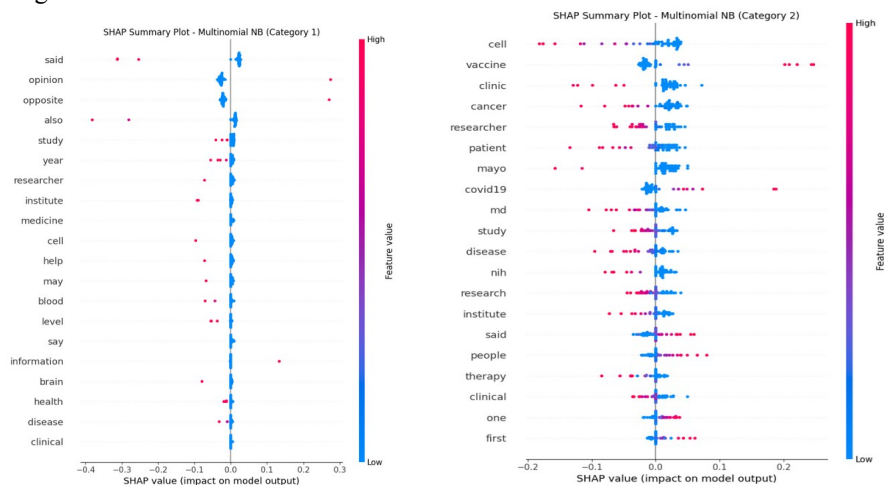
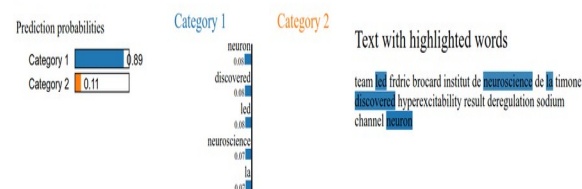
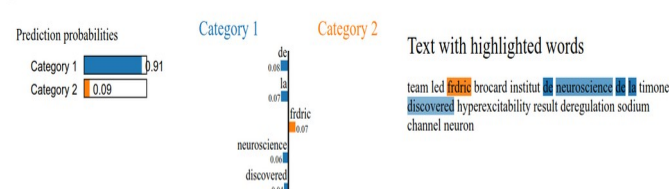


Figure 2. SHAP Plot Examples

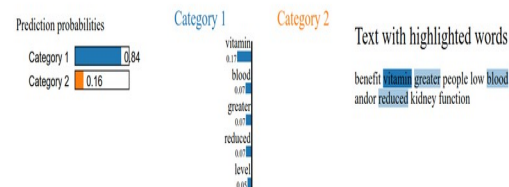
Test Example 1: team led frdric brocard institut de neuroscience de la timone discovered hyperexcitability result deregulation sodium channel neuron



Test Example 1: team led frdric brocard institut de neuroscience de la timone discovered hyperexcitability result deregulation sodium channel neuron



Test Example 2: benefit vitamin greater people low blood vitamin level andor reduced kidney function



Test Example 2: benefit vitamin greater people low blood vitamin level andor reduced kidney function

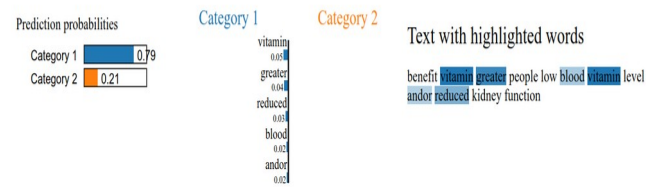


Figure 3. LIME Plot Example

5. Conclusion

Our results showed that combining data balancing, message-length categorization, and advanced models like FNN, BERT, and BioBERT led to strong improvements in detecting medical misinformation. BioBERT outperformed all other models, especially on long texts (F1 up to 0.98), while short texts remained challenging with lower F1-scores (0.68–0.75), mainly due to limited context. Classical models like Logistic Regression and SVM performed well in certain cases, especially with SMOTE, but struggled in more ambiguous categories. SHAP and LIME helped us interpret predictions, revealing that general or emotional words in short tweets often misled the models. Scientific and domain-specific terms, on the other hand, were strong indicators of credibility.

To improve short-text performance, we suggest targeted preprocessing to filter misleading terms and a refined sampling strategy, followed by re-training.

Future work

Research could focus on enhancing short-text classification by incorporating context-aware preprocessing, such as phrase disambiguation or semantic filtering. Leveraging multi-modal inputs -- including metadata and image features -- may also enrich the signal for better performance. Additionally, techniques like domain-adaptive pretraining, contrastive learning, and curriculum learning can help models distinguish subtle misinformation patterns. Evaluating models under adversarial and real-world noisy conditions would further ensure reliability in practical deployment.

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